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Citation: Review of Scientific Instruments **75**, 5106 (2004); doi: 10.1063/1.1818531
View online: http://dx.doi.org/10.1063/1.1818531
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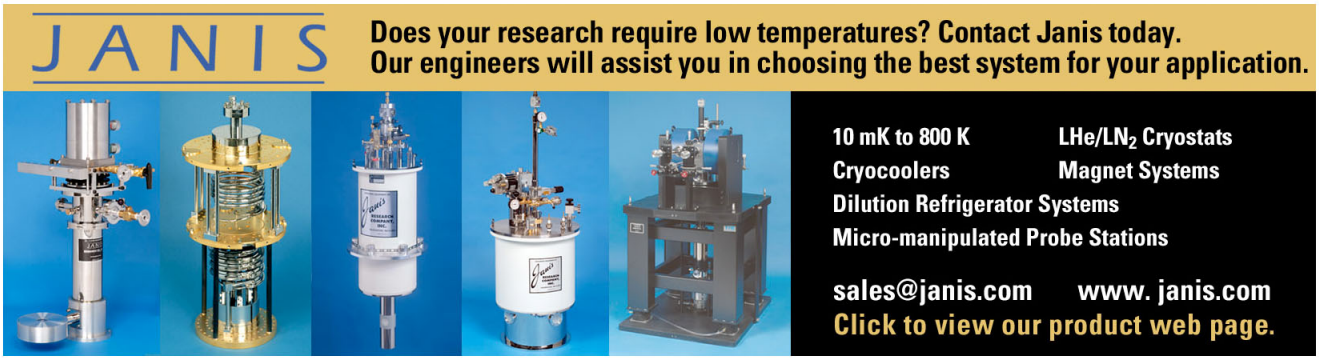
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# Optimal two-point static calibration of measurement systems with quadratic response

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(Received 29 July 2004; accepted 23 August 2004; published 10 November 2004)

Measurement devices and instruments must be calibrated after manufacture to correct for component and assembly tolerances, and periodically to correct for drift and aging effects. The number of reference inputs needed for calibration depends on the actual transfer characteristic and the desired accuracy. Often, a linear characteristic is assumed for simplicity, either for the overall input range (global linearization) or for successive input subranges (piecewise linearization). Thus, only two reference inputs are needed for each straight line. This two-point static calibration can be easily implemented in any system having some basic computation capability and allows for the correction of zero and gain errors, and of their drifts if the system is periodically calibrated. Often, the reference inputs for that calibration are the end values of the measurement range (or subrange). However, this is not always the optimal selection because the calibration error is minimal for those reference inputs only, which are not necessarily the most relevant inputs for the system being considered. This article proposes three optimization criteria for the selection of calibration points: limiting the maximal error (LME), minimizing the integral square error (ISE), and minimizing the integral absolute error (IAE). Each of these criteria needs reference inputs whose values are symmetrical with respect to the midrange input ($x_c$), have the form $x_c \pm \Delta x/(2\sqrt{n})$ when the measurand has a uniform probability distribution function, $\Delta x$ being the measurement span, and do not depend on the nonlinearity of the actual response, provided this is quadratic. The factor $n$ depends on the particular criterion selected: $n=2$ for LME, $n=3$ for ISE, and $n=4$ for IAE. These three criteria give parallel calibration lines and can also be applied to other nonlinear responses by dividing the measurement span into convenient intervals. The application of those criteria to the linearization of a type-J thermocouple illustrate their performance and advantages with respect to the customary end-point linearization ($n=1$) even for nonquadratic responses. For quadratic responses, $n=1$ yields the maximal error at the center of the input measurement range. © *2004 American Institute of Physics.* [DOI: 10.1063/1.1818531]

## I. INTRODUCTION

A common method to reduce static error in electronic measurement systems or subsystems is by periodic internal calibration. Calibration is defined by the International Standards Organization (ISO) as *the set of operations which establish, under special conditions, the relationship between values indicated by a measuring instrument or measuring system, or values represented by a material measure and the corresponding known values of a measurand.*[1] Calibration is performed by applying accurate reference inputs (measurand values) to the system or subsystem to calibrate. The corresponding measured outputs inform about the actual system response and can be used to determine the input associated to any particular output. Calibration permits the correction of zero, gain, and nonlinearity errors. There are three groups of digital methods to determine the input from the output:[2] (1) look-up tables in a read-only memory and linear interpolation, (2) calculation of the inverse function, and (3) interpolation formulas using three or more calibration points.

Look-up tables require many calibration points, whose number can be reduced by interpolating between them. However, manufacturing tolerances and drifts cannot be corrected. Calculating the inverse of the function that relates the output to the input requires us first to determine that function, which needs many reference inputs again. Storage needs are smaller than for the look-up table method, but manufacturing tolerances and drifts cannot be corrected. Interpolation formulas based on correction coefficients derived from a few internal calibration points permit the correction of manufacturing tolerances and drifts by updating those coefficients. The computation load depends on the number of points and the interpolation algorithm. Dias Pereira *et al.*[1] compare the relative performance of different interpolation and regression methods for 5, 10, 15, 20, and 25 calibration points. The three-point method described by Bolk[2] and the noniterative multipoint method proposed by Huijsing and co-workers[3] are more efficient than polynomial fitting. Linear interpolation needs only two reference points and can be easily performed by low-cost microcontrollers, but it is ob-

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viously limited to input ranges that yield an approximately linear output. Efficient piecewise linear interpolation can be implemented by linearization of the derivative of the system transfer characteristic.[4]

Whatever the interpolation method, the question about the optimal selection of the calibration points remains open. This is also true for analog linearization, which is common for sensor bridge circuits.[5] Calibration minimizes errors at the reference points but not for inputs far from them. In two-point calibration, for example, often the two reference values are a zero input and the end-of-scale input, and therefore inputs close to the center of the scale have the maximal error. For a measurand with a Gaussian probability distribution function (PDF) this would result in a large error most of the time. For a measurand with a uniform (rectangular) PDF, the error would not be uniform. Nevertheless, zero and end-of-scale inputs are so easily available that they have become the preferred reference values. A zero input, for example, means a short circuit in resistance measurements,[6] an open circuit in capacitance measurements,[7] and a ground connection in voltage measurements. Other input quantities do not have such easily available reference values, and alternative selection criteria are preferred. Fraden,[8] for example, describes a procedure for two-point thermistor calibration that involves selecting a reference temperature near the most commonly operating temperature, and the second calibrating temperature near the highest operating temperature.

Determining the optimal calibration points clearly depends on the actual system response, the measurand's PDF, which is normally unknown, and the acceptable error pattern, which depends on the application; a large error for input values that seldom occur can be acceptable if the measurement result is not used for critical decisions. We initially restrict ourselves here to systems whose actual response is quadratic, and assume an input with uniform PDF. (The Appendix deals with a pseudotriangular PDF.) Three error criteria will be considered and the respective reference inputs calculated: (A) a limited maximal error (LME) at both ends of the measurement range; (B) the absolute difference between the actual response and the ideal response is minimal for the complete input range (integral of the absolute magnitude of the error, IAE); and (C) the integral of the square of the error (ISE) is minimal.

Nonlinearity in measurement systems can result from both the sensor and from their electronic interfaces. Quadratic responses in particular, are obtained for example in semiconductor strain gages, some resistive temperature sensors, and some thermocouples (in a limited temperature range). Wheatstone bridges with a single linear sensor, and relaxation and harmonic oscillators whose frequency is inversely proportional to a linear resistive or capacitive sensor yield nonlinear outputs[9] that can be approximated by a quadratic function. Nevertheless, the method here proposed and, to some extent, the value for the reference inputs determined, can be applied to systems whose response is not quadratic, provided the measurement range or interval considered is narrow enough for a quadratic curve to approximately describe the actual response.

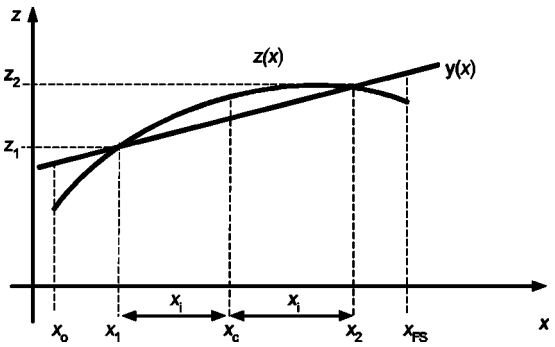



FIG. 1. Ideal response $z(x)$ and optimal calibration line $y(x)$ determined by the optimal reference inputs $x_1=x_c-x_i$ and $x_2=x_c+x_i$.

## II. OPTIMAL POINT CALCULATION FOR QUADRATIC RESPONSES

The output of a system with quadratic response is

$$z = ax^2 + bx + c, \tag{1}$$

where $x$ is the input and $a$, $b$, and $c$ are system parameters. If the ideal response is $y=Kx+y_0$, $y$ being the theoretical (ideal) output, $K$ the expected static sensitivity, and $y_0$ the output for $x=0$, we say that there is a zero (or offset) error if $c \neq y_0$, a gain error if $b \neq K$, and a nonlinearity error if $a \neq 0$. Solving for $x$ in Eq. 1 would yield the correct input from the measured $z$, but this would require us to know $a$, $b$, and $c$, and performing calculations such as a square root plus several multiplications, subtractions, and a division involving not only $a$, $b$, and $c$, but also $z$, which changes from measurement to measurement.

Calibrating with two reference inputs $x_1$ and $x_2$ would yield, respectively, $z_1$ and $z_2$ (Fig. 1). From $(x_1,z_1)$ and $(x_2,z_2)$, for any output $z$ we can calculate the estimated input $v$, as

$$v = \frac{x_2 - x_1}{z_2 - z_1}(z - z_1) + x_1. \tag{2}$$

Here, $z$ is included in a subtraction and a simple product by a calibration parameter. Hence, $v$ is easier to calculate than $x$ in Eq. (1). Ideally $v=x$, and the error would be zero. In practice, replacing $z$, $z_1$, and $z_2$ [obtained from Eq. (1)], yields

$$v = \frac{a(x^2 + x_1x_2) + bx}{a(x_1 + x_2) + b}. \tag{3}$$

Therefore, $v=x \forall x$ only when $a=0$; that is, when the response is linear. Our aim is to select $x_1$ and $x_2$ to minimize the absolute error $e=|v-x|$, as

$$e(x) = \left| \frac{a(x - x_1)(x - x_2)}{a(x_1 + x_2) + b} \right|. \tag{4}$$

Of course, $e=0$ for $x \in \{x_1,x_2\}$, but otherwise $e \neq 0$. Not surprisingly, by equating to zero the first derivative of $e$ with respect to $x$, it results that the maximal error happens at midpoint between $x_1$ and $x_2$, and its value is

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$$e_{\max} = \left| \frac{-a}{4} \frac{(x_2 - x_1)^2}{a(x_1 + x_2) + b} \right|. \quad (5)$$

This error pattern would be particularly inconvenient for measurands whose more relevant values for the application in hand were those at midinput range.

The optimal values for the reference inputs $x_1$ and $x_2$ are those that minimize the absolute error $e$ in some way. We consider here the three minimization criteria defined in Sec. I.

### A. Limited maximal error (LME)

We are interested in determining which reference inputs $x_1$ and $x_2$ yield an error for a given input $x_o$ that is $k$ times the maximal error in Eq. (5). That is,

$$\left| \frac{a(x_o - x_1)(x_o - x_2)}{a(x_1 + x_2) + b} \right| = k \left| \frac{a}{4} \frac{(x_2 - x_1)^2}{a(x_1 + x_2) + b} \right|. \quad (6)$$

Rearranging this equation yields

$$x_o^2 - x_o(x_1 + x_2) + x_1 x_2 - \frac{k(x_2 - x_1)^2}{4} = 0, \quad (7)$$

and, solving for $x_o$, we obtain

$$x_{o1} = \frac{x_1(1 + \sqrt{1+k}) + x_2(1 - \sqrt{1+k})}{2}, \quad (8)$$

$$x_{o2} = \frac{x_1(1 - \sqrt{1+k}) + x_2(1 + \sqrt{1+k})}{2}. \quad (9)$$

Solving now for the reference inputs $x_1$ and $x_2$ in Eqs. (8) and (9) yields

$$x_1 = \frac{x_{o1}(1 + \sqrt{1+k}) - x_{o2}(1 - \sqrt{1+k})}{2\sqrt{1+k}}, \quad (10)$$

$$x_2 = \frac{x_{o2}(1 + \sqrt{1+k}) - x_{o1}(1 - \sqrt{1+k})}{2\sqrt{1+k}}. \quad (11)$$

If we accept that the input values leading to the maximal absolute error be the extremes of the measurement range, i.e., $x_{o1}=x_o$ and $x_{o2}=x_{FS}$, the reference inputs $x_1$ and $x_2$ should be

$$x_1 = \frac{x_o(1 + \sqrt{1+k}) - x_{FS}(1 - \sqrt{1+k})}{2\sqrt{1+k}}, \quad (12)$$

$$x_2 = \frac{x_{FS}(1 + \sqrt{1+k}) - x_o(1 - \sqrt{1+k})}{2\sqrt{1+k}}. \quad (13)$$

If the midrange value $(x_0+x_{FS})/2$ is called $x_c$ and the input span is $\Delta x = x_{FS} - x_o$, the equations above simplify to

$$x_1 = x_c - \frac{\Delta x}{2\sqrt{1+k}}, \quad (14)$$

$$x_2 = x_c + \frac{\Delta x}{2\sqrt{1+k}}. \quad (15)$$

Therefore, the optimal reference inputs $x_1$ and $x_2$ do not depend on $a$, $b$, or $c$; that is, they are independent from the actual (unknown) system response. In addition they are symmetrical with respect to the midrange input $x_c$.

### B. Minimal integral absolute error (IAE)

An alternative criterion is that the difference between the calculated input and the actual input be minimal for the entire input range. This penalizes having relatively large errors for a broad input range. We define the integral absolute error as

$$\mathrm{IAE} = \int_{x_0}^{x_{FS}} |v - x| dx, \quad (16)$$

$$\mathrm{IAE} = \left| \frac{a}{a(x_1 + x_2) + b} \right| \int_{x_0}^{x_{FS}} |(x - x_1)(x - x_2)| dx. \quad (17)$$

To calculate this integral, we divide the input range in three intervals: $[x_0,x_1]$, $[x_1,x_2]$, and $[x_2,x_{FS}]$, so that each factor has a defined sign inside each interval. The result is

$$\mathrm{IAE} = \left| \frac{a}{a(x_1 + x_2) + b} \right| \left[ \left( \frac{x^3}{3} - \frac{x_1 + x_2}{2} x^2 + x_1 x_2 x \right)_{x_0}^{x_1} + \left( -\frac{x^3}{3} + \frac{x_1 + x_2}{2} x^2 - x_1 x_2 x \right)_{x_1}^{x_2} + \left( \frac{x^3}{3} - \frac{x_1 + x_2}{2} x^2 + x_1 x_2 x \right)_{x_2}^{x_{FS}} \right]. \quad (18)$$

To simplify the expressions we define a new variable $x_i$ such that $x_1 = x_c - x_i$ and $x_2 = x_c + x_i$. That is, $x_i$ is the distance from the center of the input range $(x_c)$ to each reference input. Equation (18) leads then to

$$\mathrm{IAE} = \left| \frac{a}{2ax_c + b} \right| \left[ \frac{2}{3}(x_c - x_i)^3 - 2x_c(x_c - x_i)^2 + 2(x_c^2 - x_i^2)(x_c - x_i) - \frac{2}{3}(x_c + x_i)^3 + 2x_c(x_c + x_i)^2 - 2(x_c^2 - x_i^2)(x_c + x_i) + \frac{x_{FS}^3}{3} - \frac{x_0^3}{3} - x_c(x_{FS}^2 - x_0^2) + (x_c^2 - x_i^2)(x_{FS} - x_0) \right]. \quad (19)$$

The $x_i$ value that minimizes IAE is obtained by setting $d\mathrm{IAE}/dx_i = 0$, which leads to

$$4x_i^2 - x_i(x_{FS} - x_0) = 0, \quad (20)$$

whose two solutions are $x_i = 0$ and

$$x_i = \frac{x_{FS} - x_0}{4}. \quad (21)$$

Therefore, the optimal reference inputs according to this criterion are

$$x_1 = x_c - \frac{\Delta x}{4}, \quad (22)$$

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$$x_2 = x_c + \frac{\Delta x}{4}. \tag{23}$$

These values do not depend on $a$, $b$, or $c$, either, and they are also symmetrical with respect to $x_c$.

### C. Minimal integral square error (ISE)

The IAE criterion weights large and small errors the same. If a few too large errors are deemed more unsatisfactory than many smaller errors, we can penalize large errors by using the integral square error as

$$\mathrm{ISE} = \int_{x_0}^{x_{\mathrm{FS}}} (v - x)^2 dx, \tag{24}$$

$$\mathrm{ISE} = \left[\frac{a}{a(x_1 + x_2) + b}\right]^2 \int_{x_0}^{x_{\mathrm{FS}}} (x - x_1)^2 (x - x_2)^2 dx, \tag{25}$$

whose result is

$$\mathrm{ISE} = \left[\frac{a}{a(x_1 + x_2) + b}\right]^2 \left[\frac{1}{5}(x_{\mathrm{FS}}^5 - x_0^5) - x_c(x_{\mathrm{FS}}^4 - x_0^4) + \frac{2}{3}(x_{\mathrm{FS}}^3 - x_0^3)(3x_c^2 - x_i^2) - 2x_c(x_{\mathrm{FS}}^2 - x_0^2)(x_c^2 - x_i^2) + (x_{\mathrm{FS}} - x_0)(x_c^2 - x_i^2)^2\right]. \tag{26}$$

The $x_i$ value that minimizes ISE is obtained by setting $d\mathrm{ISE}/dx_i = 0$. This leads to

$$3(x_{\mathrm{FS}} - x_0)x_i^3 - [(x_{\mathrm{FS}}^3 - x_0^3) - 3x_c(x_{\mathrm{FS}}^2 - x_0^2) + 3x_c^2(x_{\mathrm{FS}} - x_0)]x_i = 0, \tag{27}$$

whose solutions are $x_i = 0$ and

$$x_i = \frac{x_{\mathrm{FS}} - x_0}{2\sqrt{3}} = \frac{\Delta x}{2\sqrt{3}}. \tag{28}$$

The optimal calibration points are then

$$x_1 = x_c - \frac{\Delta x}{2\sqrt{3}}, \tag{29}$$

$$x_2 = x_c + \frac{\Delta x}{2\sqrt{3}}. \tag{30}$$

Once again, these two points do not depend on the parameters $a$, $b$, and $c$ of the actual quadratic response, and are symmetrical with respect to $x_c$.

It also turns out that the optimal calibration lines derived from each of these three criteria are parallel to the straight line joining $x_0$ and $x_{\mathrm{FS}}$. To demonstrate this assertion, let us consider the straight line joining the points corresponding to the range ends $x_0$ and $x_{\mathrm{FS}}$. Because they belong to a quadratic curve, those points are $(x_0, ax_0^2 + bx_0 + c)$ and $(x_{\mathrm{FS}}, ax_{\mathrm{FS}}^2 + bx_{\mathrm{FS}} + c)$. The slope of that straight line is $m = a(x_{\mathrm{FS}} + x_0) + b$. Because the optimal calibration points for each of the three criteria here proposed fulfill the condition $x1 + x2 = x0 + x\mathrm{FS}$, $m$ is the same for the three criteria.

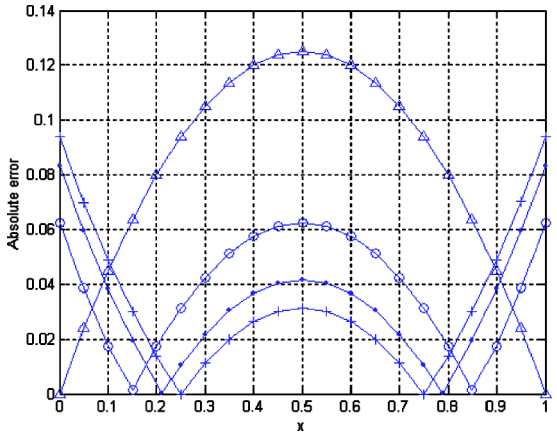



FIG. 2. (Color online) Absolute errors between the actual quadratic response in Eq. (31) and the calibration lines corresponding to limited maximal error (LME, criterion A) (circles), integral absolute error (IAE, criterion B) (crosses), integral square error (ISE, criterion C) (dots), and the end-line calibration (triangles).

## III. NUMERICAL EXAMPLE AND DISCUSSION

Let us consider a system whose actual response is

$$z = -0.5x^2 + x + 1 \tag{31}$$

and input range of [0, 1]; that is, $x_0 = 0$ and $x_{\mathrm{FS}} = 1$. Therefore, it is a system with a strong nonlinearity. If the reference inputs are accurate to 0.05, the respective calibration lines corresponding to the three proposed criteria are

$$y_{\mathrm{A}} = 0.5x + 1.06, \tag{32}$$

$$y_{\mathrm{B}} = 0.5x + 1.09, \tag{33}$$

$$y_{\mathrm{C}} = 0.5x + 1.08. \tag{34}$$

Therefore, the three calibration lines are parallel, as predicted.

Figure 2 shows the absolute error between the actual response and the four calibration lines corresponding to the three criteria proposed and the straight line through the end points $(x_0, z_0)$ and $(x_{\mathrm{FS}}, z_{\mathrm{FS}})$. When applying the LME criterion (A), we have assumed $k = 1$, which implies an error at the ends of the measurement range equal to the maximal error. The error is always minimal at the calibration points, as expected, but the IAE criterion yields less error in the center of the input range than the LME criterion, at the expense of a larger error for extreme inputs. The end-line calibration gives the maximal error in the center of the measurement range and the ISE criterion (C) yields errors in between those for criteria A and B.

These criteria for selecting reference inputs can be applied to internal calibration whenever reference values can be incorporated into the system; for example, for resistance, capacitance, and dc voltage and current measurement. They can also be applied to external calibration.

The reference inputs calculated above are optimal only for systems with quadratic response. Nevertheless, for other nonlinear responses we can divide the input range in several intervals and apply the proposed method to each interval. Similarly, if the resulting error when using two-point calibration for a quadratic response is too large for the application

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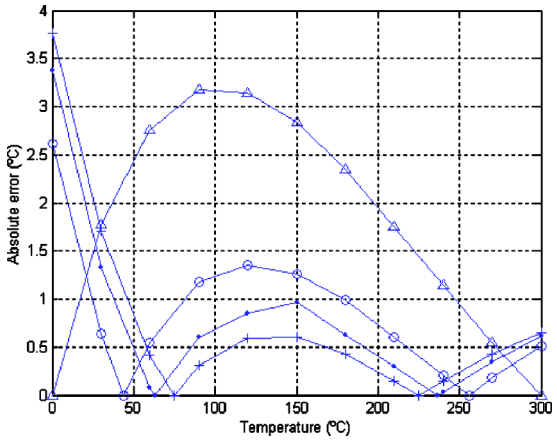



FIG. 3. (Color online) Absolute errors for a type-J thermocouple when applying the three linearization criteria proposed: limited maximal error (LME, criterion A) (circles), integral absolute error (IAE, criterion B) (crosses), integral square error (ISE, criterion C) (dots), and the end-line calibration (triangles).

in hand, the input range can be divided into several intervals and the reference inputs calculated for each interval.

## IV. APPLICATION

The proposed method has been applied to linearize the response of a type-J (Fe-CuNi) thermocouple, which has a nonlinear (and nonquadratic) characteristic, in the range from 0 to 300 °C. Figure 3 shows the absolute errors for an end-point straight line and for the three linearization criteria considered. The calibration temperatures were 44 and 256 °C for criterion A (LME), 75 and 225 °C for criterion B (IAE), and 63 and 237 °C for criterion C (ISE). The end-point line yields zero errors at the two end points, but large errors at midrange. Criterion A yields smaller errors at midrange than the end-point line, but also nonzero errors at the range ends. However, they are smaller than errors at midrange for the end-point line. Criterion B yields yet smaller errors at midrange than criterion A, but at the cost of larger errors at the range ends, as predicted. Criterion C yields intermediate errors at midrange and at range ends. None of the proposed criteria yields symmetrical errors around the midrange input (150 °C), probably because the actual response of a type-J thermocouple is nonquadratic.

To reduce errors in Fig. 3, the measurement range can be divided into two intervals; for example, 0 to 150 °C and 150 to 300 °C. To select the calibration points, we must take into account that, because the actual thermocouple characteristic is not quadratic, the absolute error is different for each end temperature. Therefore, directly selecting as calibration temperatures those determined by Eqs. (14), (15), (22), and (23), or (29) and (30) would produce two different results for 150 °C, depending on which of the two straight lines (one for each interval) were used. A possible solution is

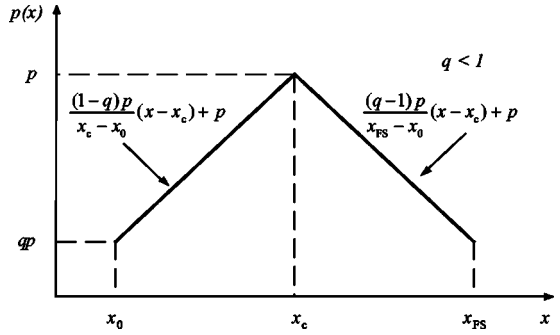



FIG. 4. Pseudotriangular probability distribution function.

to calibrate at 150 °C and to determine the additional calibration temperature at each interval by respectively using Eqs. (14) and (22) or (29) for 0 to 150 °C, and Eqs. (15) and (23), or (30) for 150 to 300 °C. Alternatively, because the errors are here larger at the low end of the range, the two intervals could be, say, 0 to 110 °C and 110 to 300 °C. In any case, the maximal error will be smaller than calibrating at 0 and 300 °C.

## ACKNOWLEDGMENT

This work has been funded by the Spanish Ministry of Science and Technology, project DPI2002-00707.

## APPENDIX: OPTIMAL CALIBRATION POINTS FOR MEASURANDS WITH PSEUDOTRIANGULAR PDF

The calibration points derived above are optimal for systems with a quadratic response when the measurand has a uniform probability density function. Figure 4 shows the following pseudotriangular probability density function

$$p(x) = \begin{cases} \dfrac{(1-q)p}{x_c - x_0}(x - x_c) + p, & x_0 \leqslant x \leqslant x_c, \\ \dfrac{(q-1)p}{x_{\mathrm{FS}} - x_c}(x - x_c) + p, & x_c \leqslant x \leqslant x_{\mathrm{FS}}, \end{cases} \tag{A1}$$

which describes those situations where measurand values close to the midrange are more probable than those near the extremes. $q$ is a parameter that determines the slope of the PDF. The probability density function fulfils the condition

$$\int_{x_0}^{x_{\mathrm{FS}}} p(x) = 1, \tag{A2}$$

and the $p$ parameter is

$$p = \frac{2}{(1+q)(x_{\mathrm{FS}} - x_0)}. \tag{A3}$$

The absolute error is now

$$e'(x) = |v - x| p(x), \tag{A4}$$

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$$e'(x)=\begin{cases}\left|\dfrac{a(x-x_1)(x-x_2)}{a(x_1+x_2)+b}p\right|\left(\dfrac{1-q}{x_c-x_0}(x-x_c)+1\right), & x_0\leqslant x\leqslant x_c,\\ \left|\dfrac{a(x-x_1)(x-x_2)}{a(x_1+x_2)+b}p\right|\left(\dfrac{q-1}{x_{\mathrm{FS}}-x_c}(x-x_c)+1\right), & x_c\leqslant x\leqslant x_{\mathrm{FS}}.\end{cases} \tag{A5}$$

If $q<1$ and we apply the same condition in Eq. (6) to calculate the LME error,

$$e'(x_o)=e'(x_{\mathrm{FS}})=ke'(x_c), \tag{A6}$$

the following equation system results:

$$q(x_1-x_0)(x_2-x_0)=q(x_{\mathrm{FS}}-x_1)(x_{\mathrm{FS}}-x_2)=k(x_c-x_1)(x_2-x_c). \tag{A7}$$

The optimal calibration points for criterion A are now

$$x_1=x_c-\frac{\Delta x}{2\sqrt{1+\dfrac{k}{q}}}, \tag{A8}$$

$$x_2=x_c+\frac{\Delta x}{2\sqrt{1+\dfrac{k}{q}}}, \tag{A9}$$

which can be obtained from Eqs. (13) and (14) by replacing $k$ by $k/q$.

The new integral square error is

$$\mathrm{ISE}'=\int_{x_0}^{x_{\mathrm{FS}}}(v-x)^2p(x)dx, \tag{A10}$$

$$\mathrm{ISE}'=\left[\frac{a}{a(x_1+x_2)+b}p\right]^2\times\left(\int_{x_0}^{x_c}\frac{[(x-x_1)^2(x-x_2)^2][(1-q)x+qx_c-x_0]}{x_c-x_0}dx+\int_{x_c}^{x_{\mathrm{FS}}}\frac{[(x-x_1)^2(x-x_2)^2][(q-1)x+x_{\mathrm{FS}}-qx_c]}{x_{\mathrm{FS}}-x_c}dx\right). \tag{A11}$$

Using again $x_1=x_c-x_i$ and $x_2=x_c+x_i$, Eq. (A11) leads to

$$\mathrm{ISE}'=\frac{2}{x_c-x_0}\left[\frac{a}{a(x_1+x_2)+b}p\right]^2\times\left(\int_{x_0}^{x_c}[(x-x_c)^2-x_i^2]^2[(1-q)x+qx_c-x_0]dx\right). \tag{A12}$$

The $x_i$ value that minimizes ISE' is obtained by setting the first derivative to zero, which leads to

$$\frac{2}{x_c-x_0}\left[\frac{a}{a(x_1+x_2)+b}p\right]^2\times\left(\int_{x_0}^{x_{\mathrm{FS}}}\frac{\partial}{\partial x_i}[(x-x_c)^2-x_i^2]^2[(1-q)x+qx_c-x_0]dx\right)=0. \tag{A13}$$

The solutions are $x_i=0$ and

$$x_i=\pm\sqrt{\frac{1+3q}{2(1+q)}}\frac{\Delta x}{2\sqrt{3}}. \tag{A14}$$

Therefore, the two optimal calibration points for criterion C are

$$x_1=x_c-\sqrt{\frac{1+3q}{2(1+q)}}\frac{\Delta x}{2\sqrt{3}}=x_c-\frac{\Delta x}{2\sqrt{\dfrac{6(1+q)}{1+3q}}}, \tag{A15}$$

$$x_2=x_c+\sqrt{\frac{1+3q}{2(1+q)}}\frac{\Delta x}{2\sqrt{3}}=x_c+\frac{\Delta x}{2\sqrt{\dfrac{6(1+q)}{1+3q}}}. \tag{A16}$$

Equations (29) and (30) are particular cases of these two equations when $q=1$.

Criterion B (IAE) does not lead to any compact expression for the optimal reference values. Nevertheless, by comparing the expressions for a uniform PDF with those for a pseudotriangular PDF, we postulate that the ranges $x_1=x_c-\Delta x/2\sqrt{n}$ and $x_2=x_c+\Delta x/2\sqrt{n}$, $n\in[2,4]$, include the optimal reference points for different criteria and PDFs.


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